



Introduction to Database System

Lab Task Chapter 3

Submitted by

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1). Write a query to display the system date. Label the column as Date.

Note: If your database is remotely located in a different time zone, the output will be the date for the operating system on which the database resides

```
SQL> select sysdate as "Date" from dual;

Date
-----
16-MAR-25

SQL>
```

2).The HR department needs a report to display the employee number, last name, salary, and salary increased by 15.5% (expressed as a whole number) for each employee. Label the column New Salary. Save your SQL statement in a file named lab_03_02.sql

EMPLOYEE_ID	LAST_NAME	SALARY	new salary
100	King	24000	27720
101	Kochhar	17000	19635
102	De Haan	17000	19635
103	Hunold	9000	10395
200	Whalen	4400	5082
201	Hartstein	13000	15015
202	Fay	6000	6930
203	Mavris	6500	7508
204	Baer	10000	11550
205	Higgins	12008	13869
206	Gietz	8300	9587

107 rows selected.

```
SQL> save lab_03_02.sql
Created file lab_03_02.sql
SQL>
```

3). Run your query in the lab_03_02.sql file

```
SQL> @lab_03_02.sql;
```

EMPLOYEE_ID	LAST_NAME	SALARY	new salary
100	King	24000	27720
101	Kochhar	17000	19635
102	De Haan	17000	19635
103	Hunold	9000	10395
104	Ernst	6000	6930
105	Austin	4800	5544
106	Pataballa	4800	5544
107	Lorentz	4200	4851
108	Greenberg	12008	13869
109	Faviet	9000	10395
110	Chen	8200	9471

```

      205 Higgins      12008      13869
      206 Gietz        8300       9587

```

107 rows selected.

```
SQL> _
```

4).Modify your query lab_03_02.sql to add a column that subtracts the old salary from the new salary. Label the column Increase. Save contents of the file as lab_03_04.sql. Run the revised query

```

SP2-0510: unable to open file 'lab_03_04.sql'
SQL> save lab_03_02.sql append
Appended file to lab_03_02.sql
SQL> ,
SP2-0042: unknown command "," - rest of line ignored.
SQL> save lab_03_02.sql append
Appended file to lab_03_02.sql
SQL> Select employee_ID, last_name, salary,
      2 round(salary*1.155) as "new salary",
      3 round((salary*1.155)-salary)as "Increase"
      4 from employees;

```

EMPLOYEE_ID	LAST_NAME	SALARY	new salary	Increase
100	King	24000	27720	3720
101	Kochhar	17000	19635	2635
102	De Haan	17000	19635	2635
103	Hunold	9000	10395	1395
104	Ernst	6000	6930	930
105	Austin	4800	5544	744
106	Pataballa	4800	5544	744
107	Lorentz	4200	4851	651
108	Greenberg	12008	13869	1861
109	Faviet	9000	10395	1395
110	Chen	8200	9471	1271

```

      205 Higgins      12008      13869      1861
      206 Gietz        8300       9587      1287

```

107 rows selected.

```

SQL> list
      1 Select employee_ID, last_name, salary,
      2 round(salary*1.155) as "new salary",
      3 round((salary*1.155)-salary)as "Increase"
      4* from employees
SQL> _

```

5. Write a query that displays the last name (with the first letter in uppercase and all the other letters in lowercase) and the length of the last name for all employees whose name starts with the letters “J,” “A,” or “M.” Give each column an appropriate label. Sort the results by the employees’ last names. Rewrite the query so that the user is prompted to enter a letter that the last name starts with. For example, if the user enters “H” (capitalized) when prompted for a letter, then the output should show all employees whose last name starts with the letter “H.”

```
SQL> SELECT INITCAP(LAST_NAME) AS "Name",
2  LENGTH(LAST_NAME) AS "Length" FROM EMPLOYEES WHERE LAST_NAME LIKE 'J%' OR
3  LAST_NAME LIKE 'A%' OR LAST_NAME LIKE 'M%' ORDER BY LAST_NAME;
```

Name	Length
Abel	4
Ande	4
Atkinson	8
Austin	6
Johnson	7
Jones	5
Mallin	6
Markle	6
Marlow	6
Marvins	7
Matos	5
Mavris	6
Mccain	6
Mcewen	6
Mikkilineni	11
Mourgos	7

16 rows selected.

```
SQL>
```

“H” Modify the query such that the case of the entered letter does not affect the output. The entered letter must be capitalized before being processed by the SELECT query.

```
SQL> SELECT INITCAP(LAST_NAME) AS "Name",
2  LENGTH(LAST_NAME) AS "Length" FROM EMPLOYEES WHERE UPPER(LAST_NAME)
3  LIKE UPPER('&START_LETTER%') ORDER BY LAST_NAME;
Enter value for start_letter: H
old 3: LIKE UPPER('&START_LETTER%') ORDER BY LAST_NAME
new 3: LIKE UPPER('H%') ORDER BY LAST_NAME
```

Name	Length
Hall	4
Hartstein	9
Higgins	7
Himuro	6
Hunold	6
Hutton	6

6 rows selected.

```
SQL> _
```

6. The HR department wants to find the duration of employment for each employee. For each employee, display the last name and calculate the number of months between today and the date on which the employee was hired. Label the column as MONTHS_WORKED. Order your results by the number of months employed. Round the number of months up to the closest whole number.

Note: Because this query depends on the date when it was executed, the values in the MONTHS_WORKED column will differ for you.

```
SQL> SELECT LAST_NAME,CEIL(MONTHS_BETWEEN(SYSDATE, HIRE_DATE)) AS "MONTHS_WORKED"  
2 FROM EMPLOYEES ORDER BY "MONTHS_WORKED";
```

LAST_NAME	MONTHS_WORKED
Banda	203
Kumar	203
Ande	204
Markle	205
Lee	205
Philtanker	206
Zlotkey	206
Marvins	206
Geoni	206

Raphaely	268
Faviet	271
Greenberg	271
Gietz	274
Mavris	274
Baer	274
Higgins	274
De Haan	291

107 rows selected.

```
SQL>
```